

Saadat Rafid Ahmed

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ACADEMIC PROFILE

Lecturer and researcher in Computer Science working on sequence models, learning under label scarcity, and the reliability of transfer-learned representations. Currently teaching Data Science, Machine Learning, NLP, and Numerical Methods. Concurrently leading an ML team that runs retrieval and extraction pipelines over noisy, mixed-script document collections at national scale, serving more than 155,000 students. Thesis research established the fragility of transfer-learned sequence classifiers under structured, character-level perturbation. Seeking a fully funded, thesis-based master's in bioinformatics and computational genomics, applying these sequence-modeling and benchmarking methods to biological sequence data.

RESEARCH INTERESTS

Current: Sequence Modeling and Representation Learning · Machine Learning under Label Scarcity · Robustness of Transfer-Learned Models · Open-Source Benchmarks and Tooling · Retrieval over Large Technical Corpora · NLP for Low-Resource Languages

Intended direction: Bioinformatics and computational genomics – carrying sequence models, transfer-learning diagnostics, and benchmark tooling across from low-resource natural language to biological sequence, where labels are similarly scarce, noise is structured rather than random, and silent model failure is similarly hard to detect.

EDUCATION

BRAC University

Dhaka, Bangladesh

B.Sc. in Computer Science and Engineering

Jan. 2020 – Dec. 2023

- CGPA 4.00 / 4.00. 100% merit-based scholarship. Vice-Chancellor's Award.
- Thesis: *destroR: Attacking Transfer Models with Obfuscated Examples to Discard Perplexity*. Advisor: Dr. Farig Yousuf Sadeque, Associate Professor, BRAC University.

RESEARCH EXPERIENCE

Thesis Research, Robustness of Transfer-Learned Sequence Classifiers

2022 – 2023

BRAC University. Advisor: Dr. Farig Yousuf Sadeque

Dhaka, Bangladesh

- Studied how transfer-learned classifiers behave under small, structured edits to the input sequence, in a domain where the annotated data behind established robustness results does not exist.
- Devised three perturbation strategies operating at the character and subword level: homoglyph substitution over visually confusable symbols, suffix perturbation against agglutinative morphology, and meaning-preserving token replacement.
- Achieved 18–40% F1 score degradation across BERT-based and LSTM-based classifiers, establishing that multilingual pretraining transfers task accuracy far more readily than it transfers robustness.
- First author; released as [arXiv:2511.11309](https://arxiv.org/abs/2511.11309) and under review at an international venue (see Previous Research Work, item 1).

Retrieval and Extraction Pipelines over Noisy Document Collections

Nov. 2024 – Present

Research strand of ML Team Lead role, Innospace Infotech Ltd.

Dhaka, Bangladesh

- Designed hierarchical retrieval combining dense embeddings with BM25 reranking over a national curriculum corpus, supporting context-grounded multi-turn query answering.
- Built an automated validation pipeline using multi-hop reasoning to check retrieved evidence against 170,000+ questions, catching silent retrieval failures that accuracy alone does not surface.
- Built a custom OCR stage for degraded scanned sources and a T5 pipeline for mixed-script extraction: a noisy-signal-to-discrete-symbol problem where off-the-shelf pipelines fail.

Thesis Supervision, Six Undergraduate Groups

May 2024 – Present

Department of CSE, BRAC University

Dhaka, Bangladesh

- Track A, Information Extraction from Specialist Corpora: retrieval, extraction, and automated summarization over domain-specific technical documents.
- Track B, User Behavior and Update Systems: ML-based engagement modeling, feedback pipelines, and behavioral analytics over longitudinal platform data.

TEACHING EXPERIENCE

Lecturer, Department of CSE

May 2024 – Present

BRAC University

Dhaka, Bangladesh

- Course co-designer for CSE437: Data Science.
- Sole course designer for CSE440 Natural Language Processing II Labs; developed the full syllabus, assignment plans, and lab content from scratch.
- Teaching over 300 students per semester across CSE437 (Data Science), CSE422 (Introduction to Machine Learning / Artificial Intelligence), CSE330 (Numerical Methods), and CSE440 (NLP II Labs)

Undergraduate Teaching Assistant

Oct. 2022 – Dec. 2023

BRAC University

Dhaka, Bangladesh

- Conducted lab sessions and led study groups for over 300 students in CSE220 (Data Structures), MAT215 (Mathematics for Machine Learning and Signal Processing), and PHY112 (Principles of Physics II).

INDUSTRY EXPERIENCE

Consultant, Data Architecture and Applied AI

Mar. 2026 – Aug. 2026

OneICT (client: National AgriCare Group)

Dhaka, Bangladesh

- Re-architected the enterprise ERP system serving roughly 10,000 internal users across the group: ETL pipelines, data migration, backup strategy, and server provisioning.
- Fine-tuned Arctic-Text2SQL-R1 on the company database schema for natural-language analytics extraction over production ERP data.
- Built a RAG-based multi-tool agentic analytics pipeline on Gemma 4, delivering a conversational analytics chatbot for business users.

Machine Learning Team Lead

Nov. 2024 – Present

Innospace Infotech Ltd.

Dhaka, Bangladesh

- Lead the AI/ML team building educational technology for Lecture Publications Ltd. and the Britto App, serving over 155,000 students.
- Delivered a QA bot, question splitter, and analytics platform.
- Architected the micro-service backend serving the platform, holding time-to-first-token at 5 seconds up to a knee of 3,000 concurrent users.
- Designed reproducible ETL pipelines that reduced costs by 52%; built a Transformer sequence tagger with 3× speedup; implemented longitudinal tracking that raised engagement by 23%.

Data Scientist

Mar. 2024 – May 2024

Robi Axiata Ltd.

Dhaka, Bangladesh

- Built predictive models over 47 million subjects using longitudinal transactional records spanning 170 TB.
- Developed a user-facing AI chatbot with chain-of-thought reasoning and an HR-facing RAG system for internal query resolution.

PREVIOUS RESEARCH WORK

** denotes first / corresponding author.*

Papers and Preprints

1. S. R. Ahmed*, “destroR: Attacking Transfer Models with Obfuscated Examples to Discard Perplexity,” arXiv preprint [arXiv:2511.11309](https://arxiv.org/abs/2511.11309), 2025. *Structured character- and subword-level perturbation of transfer-learned sequence classifiers.*

Benchmarks and Open-Source Tooling

- **BSenti**, a unified benchmark and toolkit for a low-resource sequence-classification domain. Open-source Python library consolidating fragmented, inconsistently formatted datasets behind one interface, with reproducible comparative evaluation across statistical (SVM, Naive Bayes, Logistic Regression) and deep learning (BiLSTM, BERT) baselines. Sole author.

Shared Task Participation

- **SemEval-2023 Task 10**, co-located with ACL 2023. Hierarchical, fine-grained sequence classification under severe class imbalance. Competing participant; ranked Top 12 worldwide out of 80+ teams, Micro F1 of 53.47%.

AWARDS AND HONORS

2020 – 2023 | 100% Merit-Based Scholarship, BRAC University

2020 – 2023 | Vice-Chancellor’s Award for perfect 4.00 CGPA

2023 | Top 12 Worldwide, SemEval 2023 Task 10 (EDOS), co-located with ACL 2023

SERVICE AND LEADERSHIP

Volunteer Web Systems Development

May 2024 – Present

Department of CSE, BRAC University

Dhaka, Bangladesh

- BRACU CSE SDS Portal: faculty management system centralizing administrative forms, resource requests, and workflows.
- Thesis Group Management System: end-to-end thesis lifecycle covering registration, advisor assignment, proposal tracking, panel scheduling, and final mark submission.
- Peer Evaluation System: anonymous faculty peer review with an LLM-based summarizer that preserves reviewer anonymity.

Lead Software Research Member

Dec. 2021 – Dec. 2023

Robotics and Intelligent Systems (RIS), BRAC University

Dhaka, Bangladesh

- Led software development for robotics projects and mentored junior members on embedded systems.

Senior Executive, IT Department

2021 – 2023

BRAC University Adventure Club

Dhaka, Bangladesh

TECHNICAL PROFICIENCY

Languages: Python, C, C++, TypeScript, JavaScript, Dart, WolframScript

ML, NLP, and RAG: Scikit-learn, HuggingFace Transformers, TensorFlow, TextAttack, NLTK, LangChain, vLLM, ChromaDB, Qdrant, BM25

Infrastructure and Data: AWS (SageMaker, Redshift), Docker, Kubernetes, Kafka, Apache Airflow, PostgreSQL, MongoDB, BigQuery, Groq

Frameworks: Flask, FastAPI, GraphQL, Flutter, CrewAI · **Spoken:** Bengali (native), English (fluent)

REFERENCES

Dr. Farig Yousuf Sadeque | Associate Professor, Department of CSE, BRAC University | Thesis Advisor | farig.sadeque@bracu.ac.bd